Exam policy:

- *The exam is 1 hour (7:30–8:30 PM) and is closed book, closed notes.*
- *You are allowed one hand-written cheat sheet of at most one side of one page. Please hand in your cheat sheet with your solutions.*
- *Justify all your answers. It is not enough to just have the right answer. You must also convince us that you got the right answer for the right reasons.*
- *No calculators.*

Each of the three questions is worth an equal number of points.

The exam questions are on the other side of this page.

Please do not turn over this page until you are told to start the exam.

Good luck!

Q. 1 (Successes and failures)**.** Consider a sequence of independent Bernoulli trials with success probability p at each trial. Let N be the trial number at which the first success occurs and let M be the trial number at which the first failure occurs.

a. Compute $\mathbf{P}\{M > 8\}$.

b. Compute $\mathbf{P}\{M < N\}$.

c. Compute $\mathbf{P}\{M = i, N = j\}$ for all $i, j \geq 1$.

Q. 2 (To tweet or not to tweet?)**.** When Donald Trump wakes up in the morning, the first thing he does is start composing a tweet. The amount of time (in hours) it takes him to compose his first tweet is exponentially distributed with rate 4.

The previous evening, Mr. Trump put in his breakfast order: Egg McMuffin or oatmeal, each with equal probability. The White House cook starts cooking breakfast as soon as Mr. Trump wakes up. Egg McMuffin takes 10 minutes to cook, but oatmeal takes 15 minutes to cook. (Oatmeal would cook faster in the microwave, but there are no microwaves in the White House kitchen for security reasons.) The breakfast order is independent of the amount of time it takes to compose a tweet.

Tweeting is exclusively a pre-breakfast activity: if breakfast arrives before the tweet has been fully composed, then no tweet is sent that day.

a. What is the probability that Mr. Trump has finished composing his tweet before breakfast arrives?

b. Suppose we see that a tweet was posted on a given day. Given this information, what is the probability that Mr. Trump had oatmeal for breakfast?

Q. 3 (Exam)**.** You are working questions on an exam (not this one). There are 10 questions. For each question, you have probability 0.2 to answer wrong (no points), probability 0.2 to get half points, and probability 0.6 to get full points. Each question is worth 10 points, and your performance on each question is independent.

a. What is your expected total score on the exam?

b. What is the expected number of answers you get wrong?

c. Suppose you get exactly one answer wrong on the exam. Given this information, which one of the 10 questions are you most likely to have answered wrong?